

Chatbot user archetypes & drop-off analysis

Cluster Segmentation · Funnel Analysis

THE CHALLENGE

Mental health NGO — chatbot for young users

- A conversational chatbot covering mental health quizzes, sexual health knowledge, and peer support services
- No visibility into how long users stayed, which features they engaged with, or where they dropped off
- Unable to prioritise product improvements without understanding user behaviour

KEY QUESTIONS

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How long do users stay in the chatbot?

?

Which features drive the most engagement?

?

Where do users drop off in the journey?

Digital Health Initiative

Cluster Analysis

THE ANALYSIS

Three User Archetypes

K-means clustering on chatbot interaction data revealed 3 distinct behavioural segments

Original data points replaced with relative Low / Medium / High labels. Structure and findings reflect the original analysis.

Casual User High volume, low intent		Steady User Core, high-quality users		Power User Small, high intent	
Users		Users		Users	
Large		Mid-size		Small	
Avg interaction steps		Avg interactions steps		Avg interactions steps	
Low		Medium		High	
Active days		Active days		Active days	
Low		Medium		High	
Primarily engaged with onboarding. Typically single-use — drops off early before exploring features.		Shows exploration behaviour but frequently drops out after a few steps beyond onboarding.		Highly active and frequently returns, but struggles to complete tasks due to system limitations.	

User Journey Map

● High drop-off (>40%)

Casual

n=4,754

Drop-off Rate



START



Onboard



Privacy

67%



Explore

18%

Ask

23%

Quiz

59%

Service

41%

Drop-off at Explore

Most users leave before discovering features

Steady

n=2,960

Drop-off Rate



START



Onboard



Privacy

26%



Explore

16%

Ask

12%

Quiz

48%

Service

36%

Drop-off at Quiz

Reaches Explore but exits at Quiz

Power

n=75

Drop-off Rate



START



Onboard



Privacy

0%



Explore

12%

Ask

13%

Quiz

37%

Service

21%

Friction at Service

Lowest drop-off across all menu options; willing to engage with even complex features.

Personalised Re-Engagement Nudges

Each archetype receives a message timed and framed to their specific drop-off behaviour

Casual: Re-enter the flow

Trigger: day 5–7 before full drop-off

- Short one-tap prompt to bring user back before fully disengaging
- Low-friction entry — no login or setup required

Bi

Hey! Just checking in — how have you been feeling lately? Tap to continue ✨

I'm okay, just busy

Steady: Deepen engagement

Trigger: after 2nd Explore visit, no Ask/Service

- Feature nudge highlighting unused sections to push beyond Explore
- Framed as a discovery invite, not a notification

Bi

Did you know you can Ask Big Sis anything anonymously? Tap to try it — no judgement, ever.

Oh, I didn't know that!

Power: Sustain loyalty

Trigger: new content or service update available

- Alert to new content and proactively surface nearby support services
- Treats them as a valued returning user

Bi

New: 3 support services near you just joined our network. Want me to show you?

Yes please!